

Output Form (OF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is well-aligned. Both benchmark and deployment evaluate text-based open-ended tutoring dialogue outputs. Automatic metrics (sBLEU, BERTScore, KF1, Uptake) [Q89, Q90, Q93] and interactive metrics (Success@k, Telling@k) [Q95, Q96] operate on text. Human evaluation uses 3-point Likert and binary scales [Q167] in standard formats. Authors transparently acknowledge automatic faithfulness metrics are insufficient [Q91, Q92] and that immediate solving success does not capture long-term learning [Q114]. The deployment context (English text-only on mobile/enterprise platforms) matches output modality completely. No literacy or accessibility issues for the target population (English-fluent professional teachers per population characteristics).

ID	Evidence
Q89	"We assess our models using the sacrebleu (Post, 2018) implementation of BLEU (sBLEU) (Papineni et al., 2002), as well as BERTScore between generated response ($u_T + 1$) and annotated response ($u^*_T + 1$) for each teacher response in the conversation." (p.7)
Q90	"Furthermore, in line with previous works (Dziri et al., 2022; Daheim et al., 2023), we report BERTScore and the token level F1 (KF1) between generated utterance and math word problem as a proxy for faithfulness." (p.7)
Q91	"However, we note that an increase in these metrics can be caused by an increase in overlap, which may also indicate more telling and can be undesirable." (p.7)
Q92	"However, finding good evaluation metrics for assessing the faithfulness of dialogue tutors remains an open problem." (p.7)
Q93	"Finally, we measure the Uptake of the generated response (Demszky et al., 2021)." (p.7)
Q95	"Success@k measures the percentage of conversations where the student reaches the correct final answer at least once within the first k turns (equivalent of % solve rate in prior work)." (p.7)
Q96	"Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns." (p.7)
Q114	"Finally, measuring a student's immediate success in solving a problem does not capture all the aspects of student learning. From a learning perspective, focusing on and measuring long-term learning is desired." (p.10)
Q167	"We use a 3-point Likert scale ranging from 1 (poor) and 3 (very good) for coherence and equitable tutoring and a binary scale for correctness." (p.18)

Strengths

- Output modality matches deployment exactly: text-based dialogue [Q89, Q95]
- Multiple complementary metrics provide multi-faceted evaluation [Q89, Q90, Q93, Q95, Q96]
- Authors transparent about metric limitations [Q91, Q92, Q114]
- Standard Likert/binary scales accessible to professional teacher evaluators [Q167]

ID	Evidence
Q89	"We assess our models using the sacrebleu (Post, 2018) implementation of BLEU (sBLEU) (Papineni et al., 2002), as well as BERTScore between generated response ($u_T + 1$) and annotated response ($u^*_T + 1$) for each teacher response in the conversation." (p.7)
Q90	"Furthermore, in line with previous works (Dziri et al., 2022; Daheim et al., 2023), we report BERTScore and the token level F1 (KF1) between generated utterance and math word problem as a proxy for faithfulness." (p.7)

Q91	"However, we note that an increase in these metrics can be caused by an increase in overlap, which may also indicate more telling and can be undesirable." (p.7)
Q92	"However, finding good evaluation metrics for assessing the faithfulness of dialogue tutors remains an open problem." (p.7)
Q93	"Finally, we measure the Uptake of the generated response (Demszky et al., 2021)." (p.7)
Q95	"Success@k measures the percentage of conversations where the student reaches the correct final answer at least once within the first k turns (equivalent of % solve rate in prior work)." (p.7)
Q96	"Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns." (p.7)
Q114	"Finally, measuring a student's immediate success in solving a problem does not capture all the aspects of student learning. From a learning perspective, focusing on and measuring long-term learning is desired." (p.10)
Q167	"We use a 3-point Likert scale ranging from 1 (poor) and 3 (very good) for coherence and equitable tutoring and a binary scale for correctness." (p.18)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — text-based open-ended dialogue output matches Indian metro deployment requirements (English text on mobile/enterprise platforms).

OF-2: Not applicable — deployment is text-only, no speech output required.

OF-3: Target population (professional credentialed teachers, English-fluent) has high literacy [population_characteristics]; no accessibility issues from output form.

OF-4: No form mismatches identified. Note that scoring-rubric content (Equitable Tutoring framing) is an OO concern, not OF — the form itself (Likert scale on text) is appropriate.

ID	Evidence
Q89	"We assess our models using the sacrebleu (Post, 2018) implementation of BLEU (sBLEU) (Papineni et al., 2002), as well as BERTScore between generated response ($u_T + 1$) and annotated response ($u^*_T + 1$) for each teacher response in the conversation." (p.7)
Q91	"However, we note that an increase in these metrics can be caused by an increase in overlap, which may also indicate more telling and can be undesirable." (p.7)
Q114	"Finally, measuring a student's immediate success in solving a problem does not capture all the aspects of student learning. From a learning perspective, focusing on and measuring long-term learning is desired." (p.10)
Q167	"We use a 3-point Likert scale ranging from 1 (poor) and 3 (very good) for coherence and equitable tutoring and a binary scale for correctness." (p.18)